

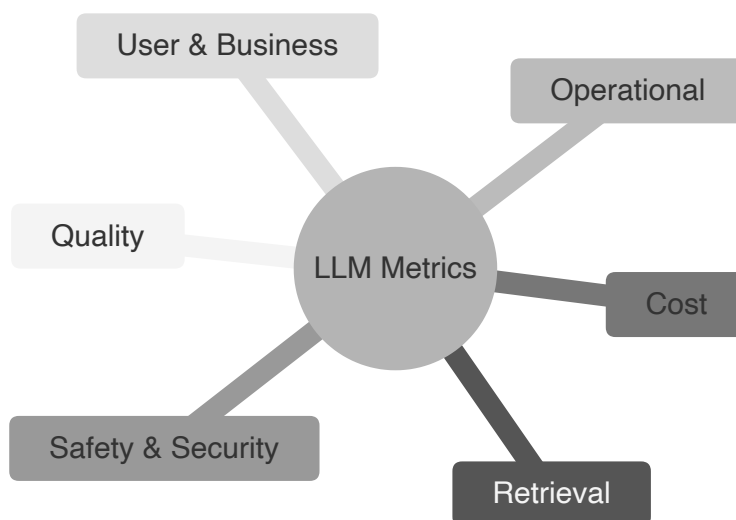
09 — LLM Metric Catalog

Part III — Observability & Metrics. A canonical, reusable catalog of metrics for any LLM system — with definitions, formulas, and how each is used as a gate.

9.1 How to use this catalog

Every metric below is defined once and referenced everywhere (eval gates, dashboards, canary thresholds, drift alarms). Adopt the naming, pick the subset relevant to your system, and wire them into [08-observability-and-opentelemetry.md](#).

Metrics are grouped into six families:



Notation: N = number of samples in the window.

9.2 Quality metrics

Metric	Definition / formula	Direction	Typical use
Faithfulness / Groundedness	fraction of answer claims supported by provided context	↑	RAG eval gate, online SLO
Answer relevance	how well the answer addresses the question (rubric/judge 1–5)	↑	Eval gate
Correctness / accuracy	agreement with reference answer (exact/semantic/judge)	↑	Extraction/classification tasks

Metric	Definition / formula	Direction	Typical use
Completeness	fraction of required elements present	↑	Structured tasks
Coherence / fluency	rubric score for readability & logic	↑	Generation quality
Hallucination rate	fraction of responses containing unsupported claims	↓	Safety SLO
Refusal rate	fraction of valid requests wrongly refused	↓	Over-blocking check
Format/schema validity	fraction of outputs passing schema validation	↑	Structured output gate

$$\text{Hallucination rate} = \frac{1}{N} \sum_{i=1}^N 1[\text{response}_i \text{ contains an unsupported claim}]$$

9.3 Retrieval metrics (RAG)

Metric	Definition / formula	Direction
Context recall	fraction of ground-truth-relevant chunks retrieved	↑
Context precision	fraction of retrieved chunks that are relevant	↑
Recall@k / Precision@k	recall/precision within top- k	↑
MRR	$\frac{1}{N} \sum_i \frac{1}{\text{rank}_i}$ (rank of first relevant)	↑
nDCG@k	rank-weighted relevance gain	↑
Citation correctness	fraction of citations pointing to supporting chunk	↑
Chunk utilization	fraction of retrieved context actually used	↑

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}_i} \quad \text{Precision@k} = \frac{|\text{relevant} \cap \text{top-}k|}{k}$$

9.4 Operational metrics

Metric	Definition	Direction	Notes
Latency p50/p95/p99	end-to-end response time	↓	Tail latency matters most
Time to first token (TTFT)	latency until first streamed token	↓	UX for streaming
Tokens per second	output throughput	↑	Perceived speed
Availability / success rate	successful ÷ total requests	↑	Core SLO
Error rate	provider/timeout/guardrail errors ÷ total	↓	Break down by cause
Fallback rate	requests served by a fallback provider	↓	Gateway health
Cache hit rate	cache hits ÷ total	↑	Cost & latency

Metric	Definition	Direction	Notes
Retry rate	retried ÷ total	↓	Cost & instability signal
Queue/concurrency saturation	in-flight ÷ capacity	↓	Capacity planning

9.5 Cost metrics (FinOps)

Metric	Definition / formula	Direction
Input/output tokens per request	mean tokens in/out	↓
Cost per request	$T_{in}p_{in} + T_{out}p_{out} + \dots$	↓
Cost per resolved request	total spend ÷ successful outcomes	↓
Cost by tenant / feature / model	attributed spend	monitor
Budget utilization	spend ÷ budget	↓
Savings from routing/caching	baseline – actual	↑

See [06-llm-finops.md](#) for definitions and enforcement.

9.6 Safety & security metrics

Metric	Definition	Direction
Guardrail block rate	requests blocked by input/output guardrails	monitor
Prompt-injection detection rate	detected injection attempts (of known set)	↑
PII leakage rate	outputs containing unredacted PII	↓ (target 0)
Toxic/unsafe output rate	outputs flagged by safety classifier	↓
Jailbreak success rate	red-team prompts that bypass controls	↓ (target 0)
Unauthorized tool-call attempts	agent calls blocked by allowlist	monitor
Policy violation rate	domain-policy breaches	↓

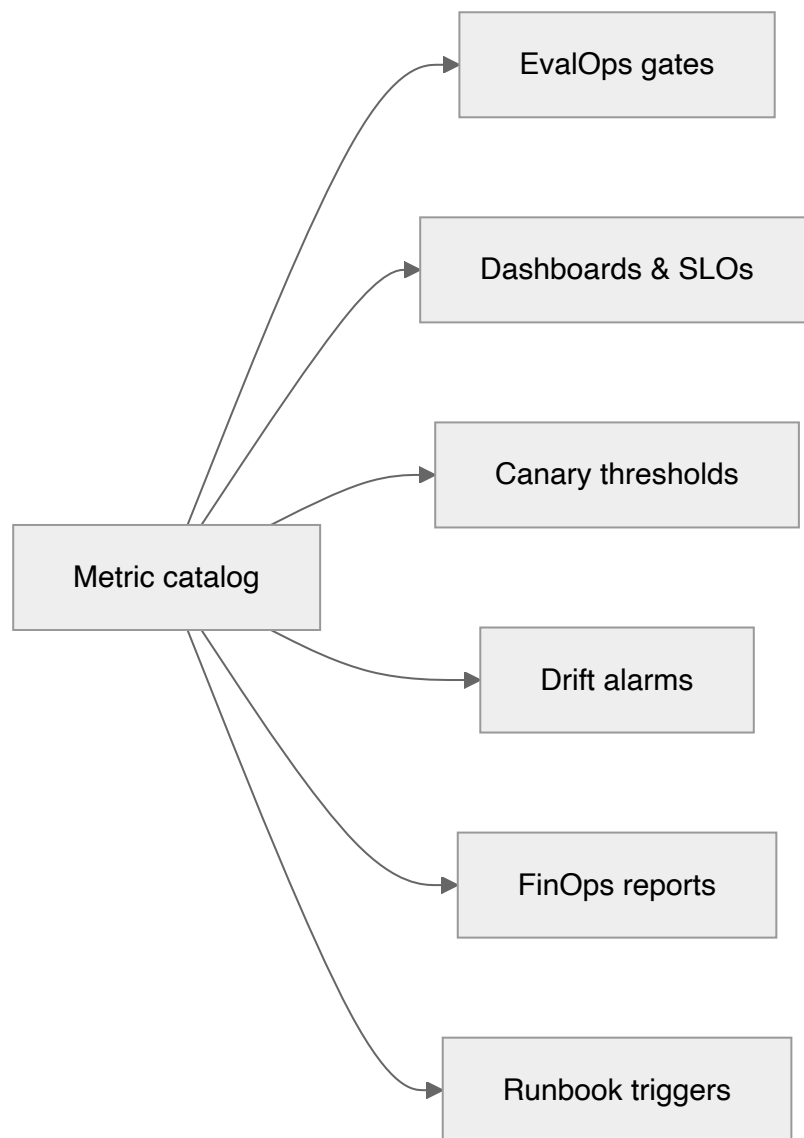
These map directly to the OWASP LLM Top 10 controls in [10-security-architecture.md](#).

9.7 User & business metrics

Metric	Definition	Direction
Thumbs-up / satisfaction rate	positive feedback ÷ rated	↑
Task success / resolution rate	tasks completed without escalation	↑

Metric	Definition	Direction
Deflection / containment rate	resolved without human handoff	↑
Edit distance	how much users edit generated output	↓
Escalation rate	handed to a human	context-dependent
Adoption / active usage	usage over time	↑

9.8 Metric wiring: one signal, many consumers



Practice. Define each metric **once** with an explicit formula, window, and owner. Reuse the same definition for the offline eval gate, the online SLO, and the canary threshold so numbers are comparable everywhere.

9.9 A minimal starter set

If you adopt nothing else, track these ten:

1. Faithfulness/groundedness (quality)
 2. Answer relevance or task success (quality)
 3. Context recall (retrieval)
 4. Latency p95 + TTFT (operational)
 5. Success/error rate (operational)
 6. Cost per resolved request (cost)
 7. Hallucination rate (safety)
 8. PII leakage rate (security — target 0)
 9. Guardrail block rate (security)
 10. User satisfaction / thumbs-up (business)
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9.10 Checklist

- ☐ Each adopted metric has a documented formula, window, direction, and owner.
 - ☐ Metrics are emitted with attribution tags (tenant, feature, model, prompt version).
 - ☐ The same metric definition feeds eval gates, SLOs, canary thresholds, and drift alarms.
 - ☐ Quality, retrieval, operational, cost, safety, and business families are all represented.
 - ☐ Targets/thresholds are set relative to a measured production baseline.
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References

See [19-sources-and-references.md](#) :

- RAGAS / DeepEval / TruLens metric definitions.
- OpenTelemetry GenAI metrics semantic conventions.
- Google SRE — SLIs/SLOs; Stanford HELM evaluation dimensions.